



Global Renewable Energy in 2025: How Far Have We Come?

For policy analysts and energy sector strategists

Source: Our World in Data (Ember 2026; Energy Institute 2025)

Fossil Fuels Still Drive Most Energy—And Most Harm

Three-quarters of GHG emissions come from fossil fuels for energy

~75%

of global greenhouse gas
emissions come from burning
fossil fuels for energy

≥5M

premature deaths per year
linked to air pollution from
fossil fuel combustion

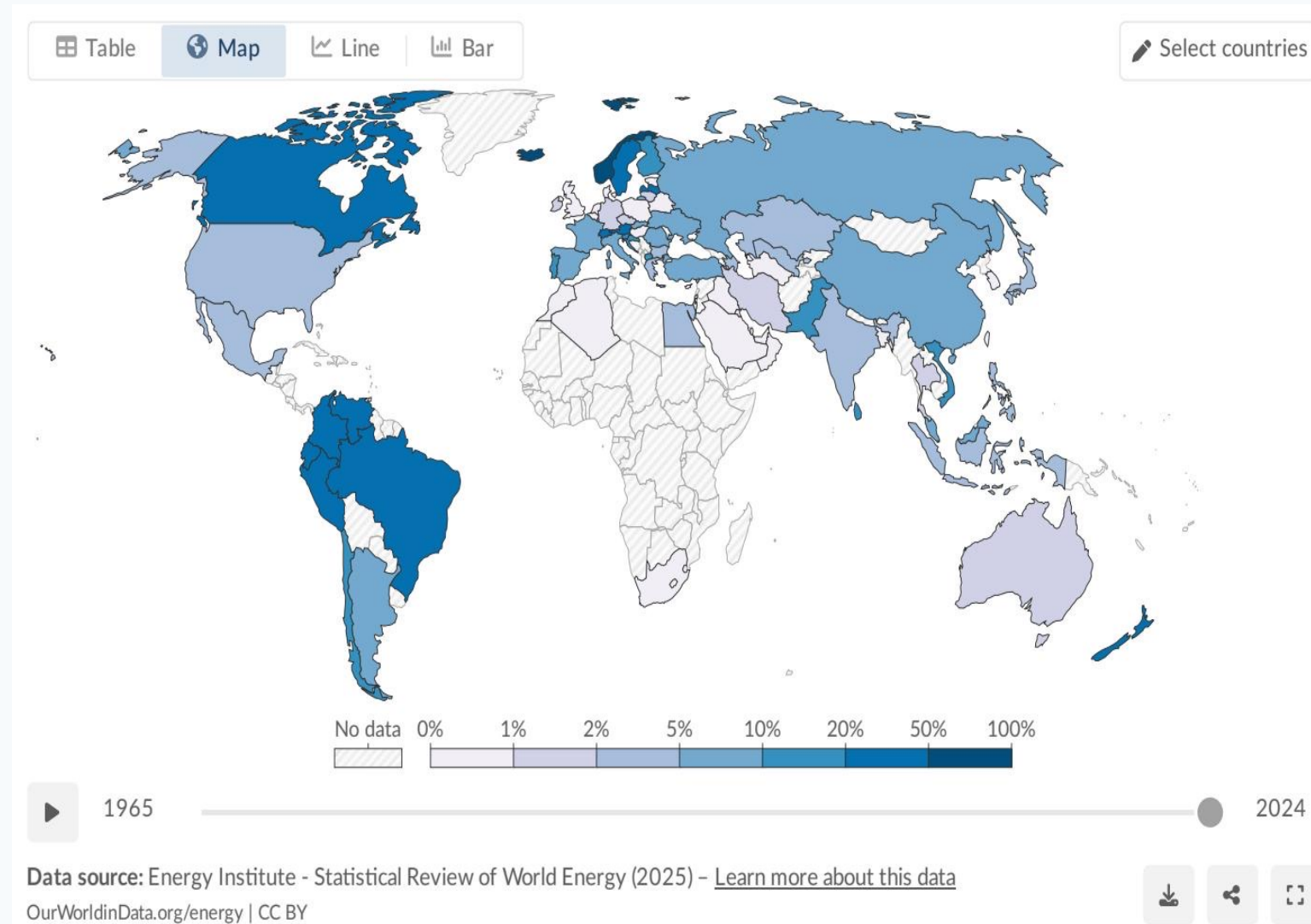
Policy Implication

Rapid transition to
low-carbon energy is
necessary for climate
and public health.

Renewables now include modern bioenergy, hydropower, wind, solar, geothermal, wave, and tidal.
Traditional biomass is typically excluded from modern renewable statistics.

Only ~14% of Global Primary Energy Is Renewable (2024)

Primary energy includes electricity, transport, and heating



~14%
one-seventh

Why this share is lower than electricity:

- Transport and heating remain more dependent on oil and gas
- Decarbonization progress is concentrated in power systems

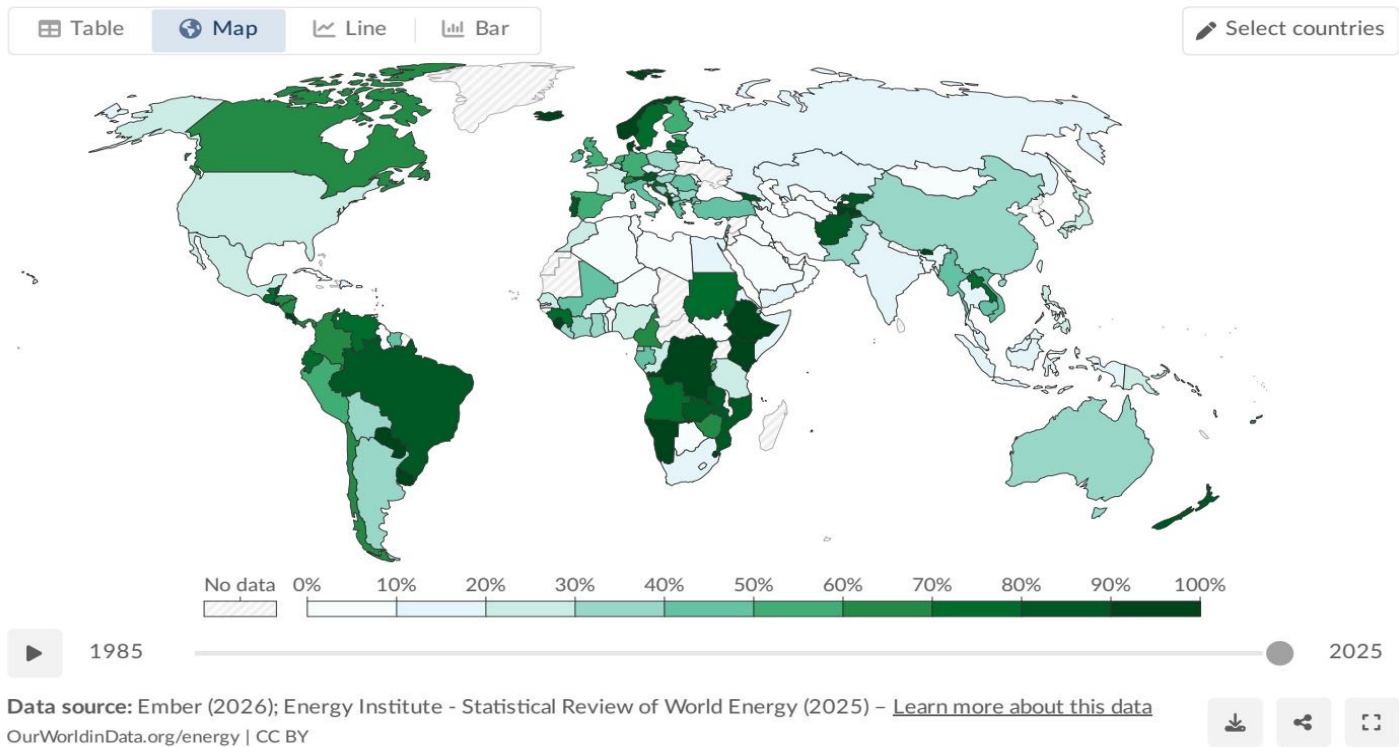
High-renewable primary energy countries include Norway, Brazil, and New Zealand; many Middle Eastern producers remain below 5%.

Global Electricity Is Now ~30% Renewable (2025)

Renewables progress is strongest in electricity generation

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

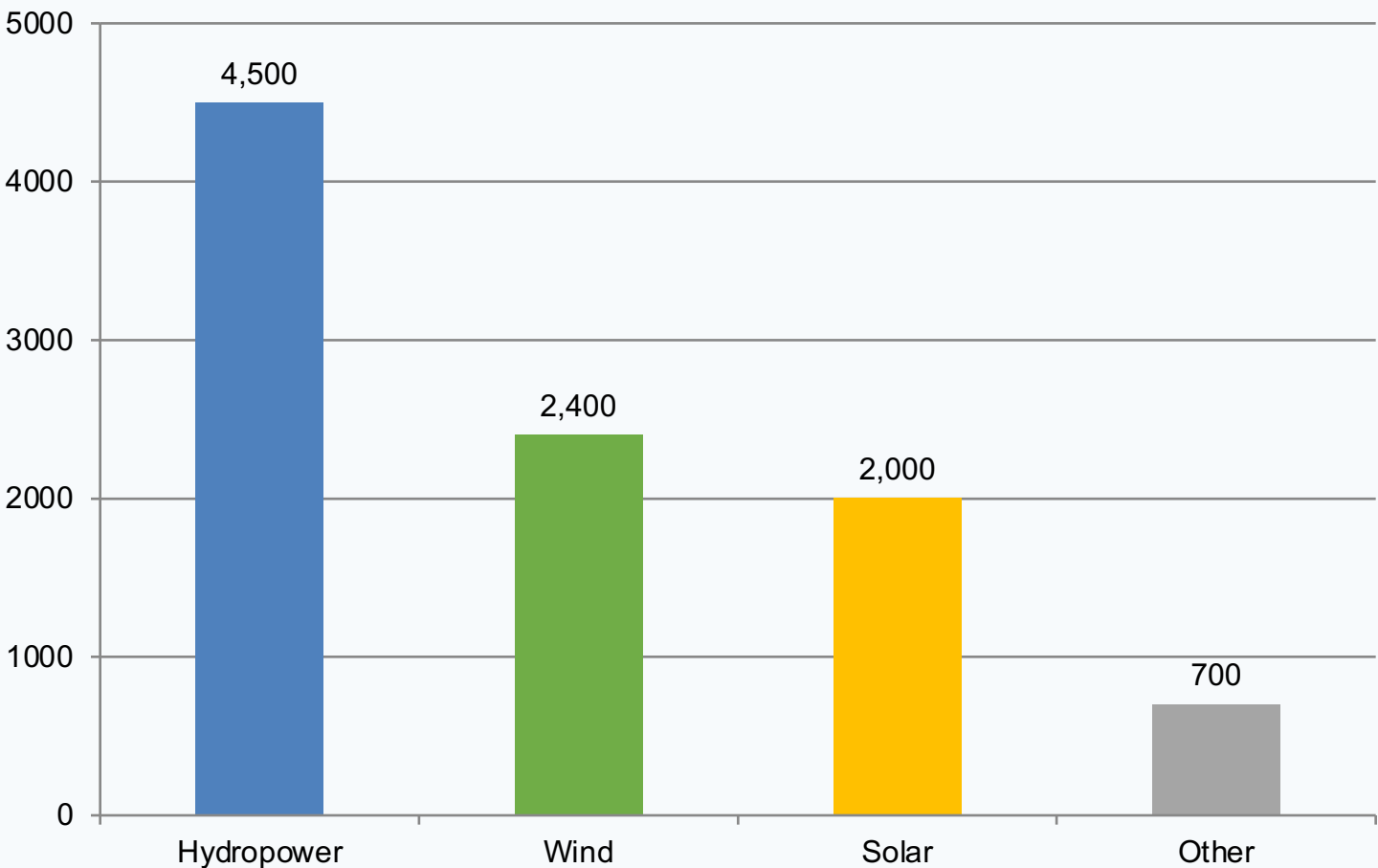


Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Gap remains stark: several African and Middle Eastern systems are below 10%, while hydro-rich systems lead above 65%.

Hydro Leads Renewable Power Output, But Wind and Solar Are Close Behind

Approx. global renewable electricity generation by technology (2024, TWh)

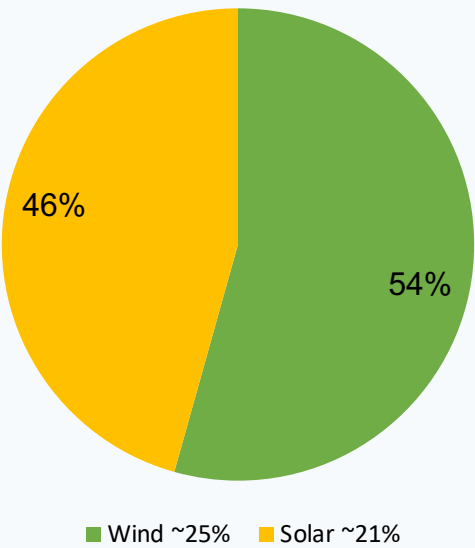
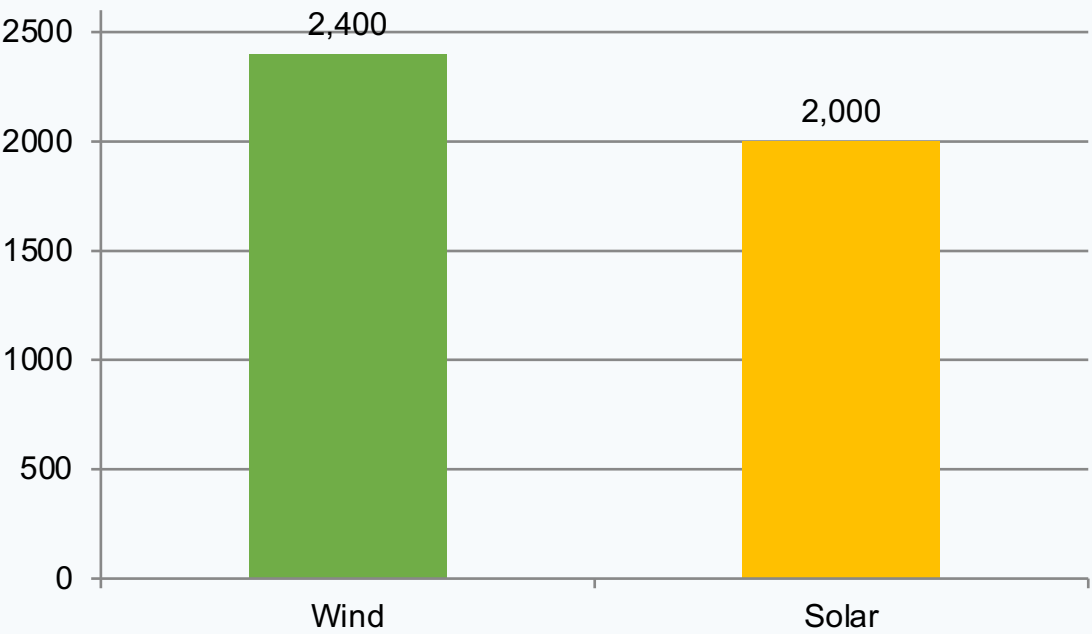


Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Wind + solar combined (~4,400 TWh) now nearly match hydropower output (~4,500 TWh).

Wind and Solar Are the Fastest-Growing Renewable Electricity Sources

Generation and share comparison from 2024 global totals



**Combined output:
~4,400 TWh**

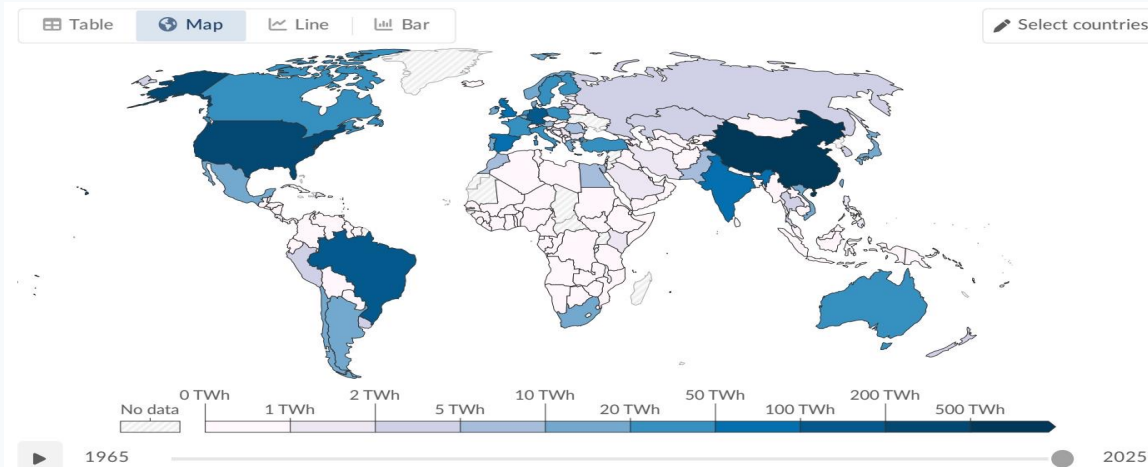
**Combined share:
~46% of renewable
electricity**

Interpretation: hydropower remains largest individually, but wind and solar together now form the main growth engine of renewable electricity expansion.

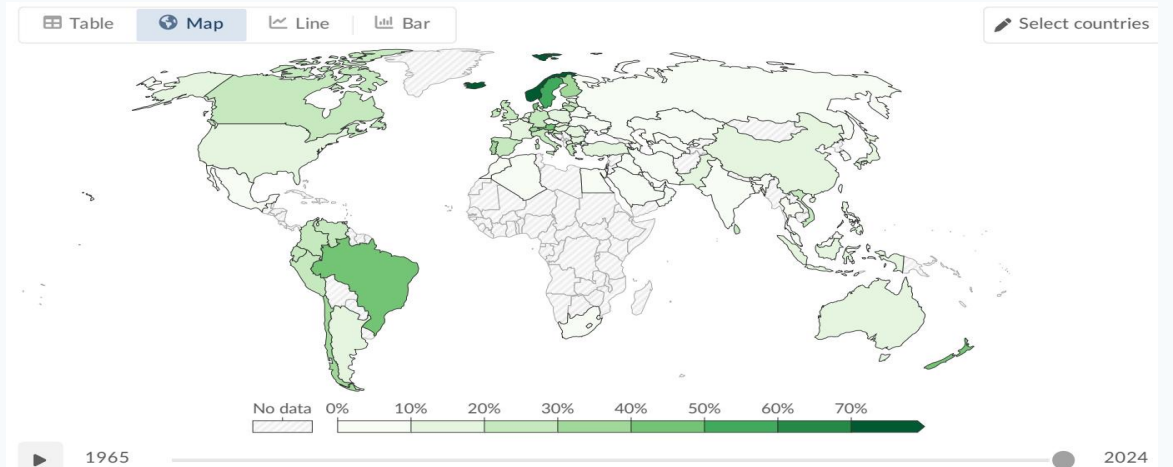
Hydropower Is Mature; Wind and Solar Drive Incremental Growth

Country patterns differ by resource endowment and policy support

Hydropower



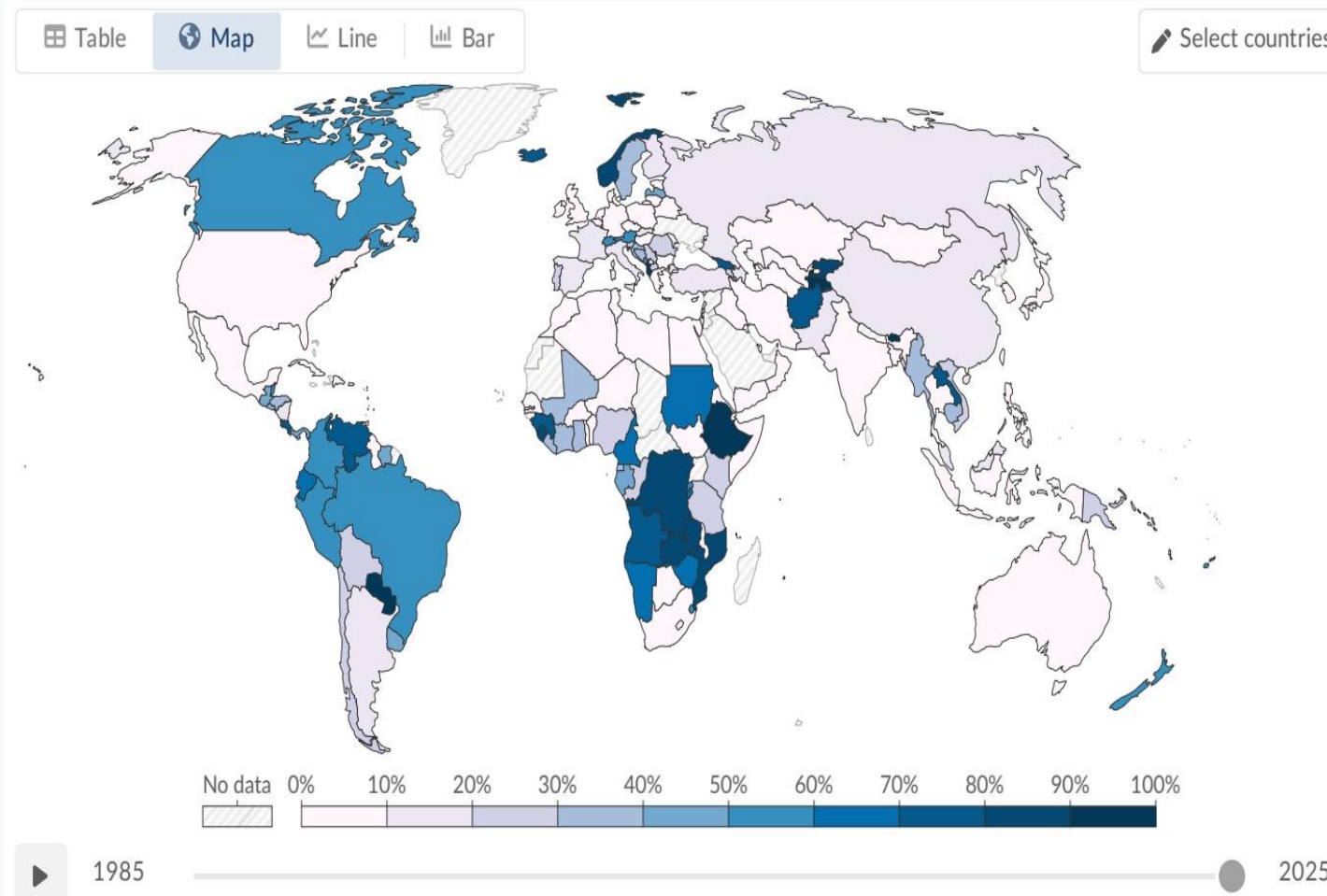
Wind



Hydropower remains the largest renewable electricity source globally, with major output from China, Brazil, and Canada. Wind has scaled rapidly across Europe, China, and the U.S., while solar is expanding fastest from a lower base due to steep cost declines and policy support.

Renewable Transition Outcomes Remain Highly Uneven Across Regions

Leaders exceed 65–98% renewable electricity while many lagging systems remain below 10%



Regional pattern summary

- Scandinavia and Brazil: very high shares, supported by hydropower
- Canada: high renewable electricity share from large hydro resources
- Denmark/Germany: strong wind and solar policy-driven scaling
- Sub-Saharan Africa: limited renewable grid infrastructure
- Middle East: persistent fossil dependence, very low renewable shares

Key Takeaways: Growth Is Strong, But Pace Must Accelerate

Renewables have moved from niche to core power supply—but not yet whole-economy dominance

1. ~14% of global primary energy and ~30% of global electricity now come from renewables.

2. Hydropower remains the largest single renewable electricity source (~47% share).

3. Wind (~25%) and solar (~21%) are the fastest-growing and now nearly match hydro combined.

4. Regional disparities are wide: leaders exceed 60–90% renewable electricity, while many markets remain below 10%.

5. Electricity transition is ahead of transport and heating—policy must accelerate system-wide decarbonization.

Forward look: To align with climate goals, renewable deployment must continue scaling faster while grids, storage, transport, and heating transition in parallel.